

33. What is a NULL Pointer? Whether it is same as an uninitialized pointer?

Ans: Null pointer is a pointer which points to nothing but uninitialized pointer may point to anywhere.

34. Are pointers integer?

Ans: No, pointers are not integers. A pointer is an address. It is a positive number.

35. What does the error 'Null Pointer Assignment' means and what causes this error?

Ans: As null pointer points to nothing so accessing a uninitialized pointer or invalid location may cause an error.

36. What is generic pointer in C?

Ans: In C void* acts as a generic pointer. When other pointer types are assigned to generic pointer,

conversions are applied automatically (implicit conversion).

37. Are the expressions arr and &arr same for an array of integers?

Ans: Yes for array of integers they are same.

38. IMP>How pointer variables are initialized?

Ans: Pointer variables are initialized by one of the following ways.

I. Static memory allocation

II. Dynamic memory allocation

39. What is static memory allocation?

Ans: Compiler allocates memory space for a declared variable. By using the address of operator, the

reserved address is obtained and this address is assigned to a pointer variable. This way of assigning pointer value to a pointer variable at compilation time is known as static memory allocation.

40. What is dynamic memory allocation?

Ans: A dynamic memory allocation uses functions such as malloc() or calloc() to get memory dynamically. If these functions are used to get memory dynamically and the values returned by these function are assigned to pointer variables, such a way of allocating memory at run time is known as dynamic memory allocation.

41. What is the purpose of realloc?

Ans: It increases or decreases the size of dynamically allocated array. The function realloc (ptr,n) uses two arguments. The first argument ptr is a pointer to a block of memory for which the size is to be altered. The second argument specifies the new size. The size may be increased or decreased. If sufficient space is not available to the old region the function may create a new region.

42. What is pointer to a pointer?

Ans: If a pointer variable points another pointer value. Such a situation is known as a pointer to a pointer.

Example:

```
int *p1,**p2,v=10;
```

```
P1=&v; p2=&p1;
```

Here p2 is a pointer to a pointer.

43. What is an array of pointers?

Ans: if the elements of an array are addresses, such an array is called an array of pointers.

44. Difference between linker and linkage?

Ans: Linker converts an object code into an executable code by linking together the necessary

built in

functions. The form and place of declaration where the variable is declared in a program determine the linkage of variable.

45. Is it possible to have negative index in an array?

Ans: Yes it is possible to index with negative value provided there are data stored in this location. Even if it is illegal to refer to the elements that are out of array bounds, the compiler will not produce error because C has no check on the bounds of an array.

46. Why is it necessary to give the size of an array in an array declaration?

Ans: When an array is declared, the compiler allocates a base address and reserves enough space in

memory for all the elements of the array. The size is required to allocate the required space and hence size must be mentioned.

47. What modular programming?

Ans: If a program is large, it is subdivided into a number of smaller programs that are called modules or subprograms. If a complex problem is solved using more modules, this approach is known as modular programming.

48. What is a function?

Ans: A large program is subdivided into a number of smaller programs or subprograms. Each subprogram

specifies one or more actions to be performed for the larger program. Such sub programs are called functions.

49. What is an argument?

Ans: An argument is an entity used to pass data from the calling to a called function.

50. What are built in functions?

Ans: The functions that are predefined and supplied along with the compiler are known as builtin

functions. They are also known as library functions.

51. Difference between formal argument and actual argument?

Ans: Formal arguments are the arguments available in the function definition. They are preceded by

their own data type. Actual arguments are available in the function call. These arguments are given

as constants or variables or expressions to pass the values to the function.

52. Is it possible to have more than one main() function in a C program ?

Ans: The function main() can appear only once. The program execution starts from main.

53. What is the difference between an enumeration and a set of pre-processor # defines?

Ans: There is hardly any difference between the two, except that #defines has a global effect (throughout the file) whereas an enumeration can have an effect local to the block if desired. Some advantages of enumeration are that the numeric values are automatically assigned whereas in #define we have to explicitly define them. A disadvantage is that we have no control over the size of enumeration variables.

54. How are Structure passing and returning implemented by the compiler?

Ans: When structures are passed as argument to functions, the entire structure is typically pushed on

the stack. To avoid this overhead many programmer often prefer to pass pointers to structure instead of actual structures. Structures are often returned from functions in a location pointed to

by an extra, compiler-supported 'hidden' argument to the function.

55. IMP>what is the similarity between a Structure, Union and enumeration?

Ans: All of them let the programmer to define new data type.

56. Can a Structure contain a Pointer to itself?

Ans: Yes such structures are called self-referential structures.

57. How can we read/write Structures from/to data files?

Ans: To write out a structure we can use fwrite() as Fwrite(&e, sizeof(e),1,fp);Where e is a structure

variable. A corresponding fread() invocation can read the structure back from file. calling fwrite() it writes out sizeof(e) bytes from the address &e. Data files written as memory images with fwrite(),however ,will not be portable, particularly if they contain floating point fields or Pointers. This is because memory layout of structures is machine and compiler dependent. Therefore, structures written as memory images cannot necessarily be read back by programs running on other machine, and this is the important concern if the data files you're writing will ever be interchanged between machines.

58. Write a program which employs Recursion?

Ans: `int fact(int n) { return n > 1 ? n * fact(n - 1) : 1; }`

59. Write a program which uses Command Line Arguments?

Ans:

```
#include
void main(int argc,char *argv[])
{
int i;
clrscr();
for(i=0;i
printf("\n%d",argv[i]);
}
```

60. Difference between array and pointer?

Ans:

Array

- 1- Array allocates space automatically
- 2- It cannot be resized
- 3- It cannot be reassigned
- 4- sizeof (arrayname) gives the number of bytes occupied by the array.

Pointer

- 1-Explicitly assigned to point to an allocated space.
- 2-It can be sized using realloc()
- 3-pointer can be reassigned.
- 4-sizeof (p) returns the number of bytes used to store the pointer variable p.

61. What do the 'c' and 'v' in argc and argv stand for?

Ans: The c in argc(argument count) stands for the number of command line argument the program is

invoked with and v in argv(argument vector) is a pointer to an array of character string that contain the arguments.

62. IMP>what are C tokens?

Ans: There are six classes of tokens: identifier, keywords, constants, string literals, operators and

other separators.

63. What are C identifiers?

Ans: These are names given to various programming element such as variables, function, arrays. It is a combination of letter, digit and underscore. It should begin with letter. Backspace is not allowed.

64. Difference between syntax vs logical error?

Ans:

Syntax Error

- 1- These involves validation of syntax of language.
- 2- compiler prints diagnostic message.

Logical Error

- 1- logical error are caused by an incorrect algorithm or by a statement mistyped in such a way that it doesn't violate syntax of language.
- 2- difficult to find.

65. What is preincrement and post increment?

Ans: ++n (pre increment) increments n before its value is used in an assignment operation or any expression containing it. n++ (post increment) does increment after the value of n is used.

66. Write a program to interchange 2 variables without using the third one.

Ans:

a ^= b; ie a=a^b

b ^= a; ie b=b^a;

a ^= b ie a=a^b;

here the numbers are converted into binary and then xor operation is performed.

You know, you're just asking "have you seen this overly clever trick that's not worth applying on

modern architectures and only really applies to integer variables?"

67. What is the maximum combined length of command line arguments including the space between adjacent arguments?

Ans: It depends on the operating system.

68. What are bit fields? What is the use of bit fields in a Structure declaration?

Ans: A bit field is a set of adjacent bits within a single implementation based storage unit that we will call a "word".

The syntax of field definition and access is based on structure.

Struct {

unsigned int k :1;

unsigned int l :1;

unsigned int m :1;

}flags;

the number following the colon represents the field width in bits. Flag is a variable that contains three bit fields.

69. What is a preprocessor, what are the advantages of preprocessor?

Ans: A preprocessor processes the source code program before it passes through the compiler.

- 1- a preprocessor involves the readability of program
- 2- It facilitates easier modification
- 3- It helps in writing portable programs
- 4- It enables easier debugging